AYUSH GEHLOT
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PROFESSIONAL EXPERIENCE

The New York Stem Cell Foundation, New York, NY |Software Engineer June 2025 – August 2025

- Partnered with R&D engineers to prototype a smart incubator system for maintaining controlled environmental conditions in cell-culture workflows.
- Developed a C-based PID control system on Raspberry Pi 5 to regulate O_2, CO_2, N_2, internal pressure, and environmental readings in real time.
- Built a touchscreen monitoring and configuration interface supporting live sensor readings, gas setpoints, calibration workflows, alarms, and data export.
- Integrated I2C sensors, PWM-controlled solenoid valves, and logging workflows to support reliable monitoring and automated gas control.

Dream Team Technologies PVT LTD., Rajasthan, India |Full Stack Engineer June 2022 – June 2023

- Supported development and day-to-day management of an ERP-based university management system used for student enrollment, attendance, assignments, payments, and administrative workflows.
- Built and enhanced responsive ERP modules using JavaScript, HTML, CSS, and frontend frameworks including React, Angular, and Vue.js.
- Implemented OAuth-based authentication and role-based access features to support secure access for students, faculty, and administrators.
- Collaborated with developers and UX/UI designers to troubleshoot user issues, improve workflows, and deliver feature updates across the platform.
- Automated end-to-end ERP UI testing with Selenium and PyTest, reducing manual testing effort by approximately 75%.

DefenseResearch & Development Organization (DRDO), Rajasthan, India| Hardware Engineer June 2021 – March 2022

- Conducted rigorous validation of VHDL-designed circuits using Model Sim, used to design & simulate digital hardware systems, assessing syntax & structure of VHDL code through its architecture & configuration, achieving 100% verification rate, leading to increased project approvals across teams.

ACADEMICPROJECTS & EXPERIENTIAL LEARNING /HACKATHON

TrueSight - Hybrid Perceptual-Hash and CLIP-Embedding Application for Image Authentication Verification January 2026 - May2026

Python, CLIP, Perceptual Hashing, Image Similarity

- Collaborated on a five-person team to build a prototype for identifying AI-generated, manipulated, and authentic images.
- Developed an image-verification workflow using perceptual hashing and CLIP-based embeddings to compare visual similarity and detect potential manipulation.
- Researched adversarial manipulation, deepfake detection, and provenance concepts to inform system design decisions.

SmartCity Dashboard -NYC Experiential Learning Lab Sponsored by Spotify, MongoDB&Kensho Technologies (1st Place) November 2025

- Developed a responsive React-based dashboard with geospatial maps, trend graphs, and forecasting visualizations for city-wide sanitation analysis.
- Collaborated with data engineers, data scientists, and business analysts to transform statistical models into interactive insights for operational decision-making.
- Integrated MongoDB Atlas to manage 100K+ sanitation and complaint records with real-time filtering and aggregation.

AI Restaurant Chatbot forCaterPal TechnologiesHackathon (Secured 1st Place) March 2025

- Developed an AI-powered restaurant chatbot using Python (Flask), JavaScript, MongoDB, and OpenAI API to deliver real-time food recommendations based on user preferences.
- Established surplus food donation feature to connect catering services with local charities to address food waste & social impact goals while integrating feedback system to track most recommended dishes & personalize suggestions based on user preferences.

Advanced Driving Assistant System using Arduino & C++ March 2023 – June 2023

- Synchronized in team of four to develop advanced driving assistant system to support obstacle detection & vehicle navigation, integrating sensors & cameras programmed with image processing to reduce road accidents.
- Built model car with motor for system testing, creating system with Arduino sensors, LIDAR, & Integrated Radar to detect obstacles with sonic waves, integrating night-vision image sensing to boost detection accuracy & GPS APIs to direct drivers around obstacles, refining prototype to achieve approximately 70% accuracy.
- Proposed system to India's Minister of Railways, leading to integration of prototype in ministry's systems to improve railway safety.

EDUCATION

Pace University | New York, NY | Master of Science (MS) in Computer Science | **Concentration**: Software Development | **GPA:** 3.57 May 2026

MBM University | Rajasthan, India | Bachelor of Engineering (BEng) in Electronics and Computer June 2023

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Java, Python, JavaScript, R, VHDL | **Frontend:** HTML5, CSS3, React.js, Vue.js, Angular, Three.js | **Backend:** Node.js, Express.js, REST APIs, .NET | **Database Management:** MySQL, NoSQL, MongoDB, PostgreSQL | **Cloud & Platforms:** AWS EC2, AWS RDS, S3, Docker, GitHub Actions, Git | **Operating Systems:** Windows, Linux, Android, Mac OS | **Software / Tools:** VS Code, Cursor, R Studio, Tableau, Jira, Git, PyTorch, Postman, Pytest, Selenium | **Libraries:** REACT, GTK, | **Protocols:** HTTP, TCP/IP, I2C, UDP, SSH, DNS, FTP, QUIC, DHCP,SM, IMAP | **Development Methodologies:** Agile, Scrum | **AI/ML:** OpenAI API, CLIP, LangChain, Prompt Engineering, RAG, CNNs, RNNs